

CSE422 Lab Project Guideline

1. Introduction:

A small introduction to the project and its aim. Describe the problem the project is aiming to solve and the motivation behind the project.

2. Dataset Description:

- Dataset Description
 - Number of features
 - Number of data points
 - Types of features are in your dataset

3. Data Analysis:

- Perform exploratory data analysis to extract some important relationships from the data.
 - [Reference: [EDA Lab CSE422](#)]

4. Dataset Pre-processing:

- Handling null and duplicate values
- Encoding categorical values
- Feature Selection
- Feature scaling
- Other feature engineering, if necessary

Note: Discuss one problem, and then write about the solutions or pre-processing techniques you have applied to solve that problem.

5. Model Training:

- You have to apply a Neural Network and at least 2 other models such as KNN, Decision Tree, Logistic Regression, Gaussian Discriminant Analysis, Naive Bayes etc.
- Treat the problem as an unsupervised learning problem and apply k-means clustering.

6. Model Evaluation:

- Bar chart showcasing prediction accuracy and f_1 score of all models
- Confusion Matrix
- Performance metrics comparison of each model
- AUC score, ROC curve

7. Conclusion

- Best model based on the model evaluation
 - Probable reasons behind getting such results

Report Template:

- Overleaf:
<https://www.overleaf.com/latex/templates/labreport-ieee-journal-paper-template/ypbfnkby syhw>
- Docs:
<https://www.ieee.org/content/dam/ieee-org/ieee/web/org/conferences/Conference-template-A4.doc>